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Methods of making sleeved hydrophilic catheter assemblies

Abstract

Methods of making sleeved hydrophilic urinary catheters.

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Background/Summary

(1) The present application is the U.S. National Stage of International Application No. PCT/US2019/032892, filed May 17, 2019, which claims the benefit and priority to U.S. Provisional Application No. 62/672,755, filed May 17, 2018, U.S. Provisional Application No. 62/699,993, filed Jul. 18, 2018, U.S. Provisional Application No. 62/739,449, filed Oct. 1, 2018, U.S. Provisional Application No. 62/770,275, filed Nov. 21, 2018, U.S. Provisional Application No. 62/821,268, filed Mar. 20, 2019, U.S. Provisional Application No. 62/821,284, filed Mar. 20, 2019, and U.S. Provisional Application No. 62/842,318, filed May 2, 2019, all of which are hereby incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure generally relates to methods of making sleeved hydrophilic catheter assemblies wherein the catheter assemblies include a catheter tube that has an activated or hydrated hydrophilic outer surface and a barrier sleeve surrounds the catheter tube wherein the sleeve is used to grasp the catheter for manipulation and insertion into the patient.

BACKGROUND

(2) It is known to coat medical devices, such as urinary catheters, with a hydrophilic coating. When the hydrophilic coating is wetted or hydrated with a hydration medium it becomes extremely lubricous. The hydration medium may be, for example, liquid or vapor water or an aqueous

solution. The lubriciousness of the hydrophilic coating eases introduction of the device into the body and aids in reducing pain and discomfort associated with such introduction.

(3) In some urinary catheter products, the user directly contacts the urinary catheter with the user's fingers to remove the catheter from the package and inserts it into the urethra. In such products there may be a disadvantage in that the handling of the catheter by the user may introduce microorganisms onto the surface of the catheter which can cause infectious problems after being introduced into the body during catheter insertion. To address this issue, manufacturers have devised systems that include a protective or barrier sleeve surrounding the catheter. In this type of product, the catheter tube is located in a barrier sleeve. The sleeve may loosely fit the diameter of the catheter so that the user may grasp the catheter tube through the sleeve to manipulate the catheter, e.g., remove the catheter from its package and advance the catheter into the urethra. In some products, the distal end of the sleeve may be attached to the drainage member of the catheter and an insertion aid may be attached to or otherwise associated with the proximal end of the sleeve.

(4) One complication of employing a sleeve over a hydrophilic catheter is how to activate or hydrate the hydrophilic surface of the catheter located within the interior cavity of the sleeve.

SUMMARY

(5) There are several aspects of the present subject matter which may be embodied separately or together in the devices and systems described and claimed below. These aspects may be employed alone or in combination with other aspects of the subject matter described herein, and the description of these aspects together is not intended to preclude the use of these aspects separately or the claiming of such aspects separately or in different combinations as set forth in the claims appended hereto.

(6) In one aspect, a method of making a urinary catheter product, wherein the product includes a sleeve defining an inner cavity and a urinary catheter having a catheter tube located within the inner cavity of the sleeve, the catheter tube having an outer hydrophilic surface, an insertion aid at the distal end of the sleeve configured for passage of the catheter tube therethrough during catheterization, the method comprising delivering a hydration medium through the insertion aid and into the interior cavity of the sleeve, wherein the hydration medium contacts the outer hydrophilic surface of the catheter tube.

(7) In another aspect, a system for delivering hydration medium into a catheter assembly. The system includes a source of hydration fluid and a nozzle in communication with the source of hydration fluid. The nozzle is configured to dock with an insertion aid of a catheter assembly and deliver hydration fluid into the catheter assembly.

(8) In another aspect, a method of making a urinary catheter product, wherein the product includes a sleeve defining an inner cavity and a urinary catheter having a catheter tube located within the inner cavity of the sleeve, the sleeve defining at least one opening or a passageway for injecting and/or withdrawing hydration medium, the method comprising delivering a hydration medium through the insertion aid and into the interior cavity of the sleeve, wherein the hydration medium contacts the outer hydrophilic surface of the catheter tube.

(9) In another aspect, a method of making a catheter product, wherein the product includes a sleeve defining an inner cavity and a urinary catheter having a catheter tube located within the inner cavity of the sleeve, the method comprising inserting a mandrel into a drainage member of the catheter tube and delivering a hydration fluid through the mandrel.

(10) In yet another aspect, a method of making a urinary catheter product, wherein the product includes a sleeve defining an inner cavity and a urinary catheter having a catheter tube located within the inner cavity of the sleeve, the catheter tube having an outer hydrophilic surface, drainage member at the proximal end of the sleeve, the method comprising delivering a hydration medium drainage member and into the interior cavity of the sleeve, wherein the hydration medium contacts the outer hydrophilic surface of the catheter tube.

(11) In another aspect, a method of making a catheter assembly that includes a catheter having a

catheter tube with an outer hydrophilic surface and a sleeve defining an interior cavity, the catheter tube being located within the interior cavity of the sleeve. The method includes injecting a hydration medium into the interior cavity of the sleeve, wherein the hydration medium is in contact with the outer hydrophilic surface of the catheter tube. The sleeve having the catheter tube therein is then placed in an outer package.

(12) In another aspect, a method of making a catheter assembly that includes a catheter having a catheter tube with an outer hydrophilic surface and a sleeve defining an interior cavity, the catheter tube being located within the interior cavity of the sleeve. The method includes applying a hydration medium to the catheter tube. The sleeve is then assembled around the catheter tube so that the catheter is located in the interior cavity of the sleeve. The sleeve having the catheter tube therein is then placed in a package.

Description

BRIEF DESCRIPTION OF FIGURES

- (1) FIG. 1 is a perspective view of a catheter assembly in accordance with the present disclosure;
- (2) FIG. 2 is a perspective view of one embodiment of an insertion aid of the assembly of FIG. 1;
- (3) FIG. 3 is a perspective view of another embodiment of an insertion aid of the assembly of FIG. 1;
- (4) FIG. 4 is a schematic view of one embodiment of a method of making a hydrophilic sleeved catheter assembly and a hydration medium delivery device in accordance with the present disclosure;
- (5) FIG. 5 is a cross-sectional view of a nozzle engaging the insertion aid shown in FIG. 2;
- (6) FIG. 6 is a cross-sectional view of a nozzle engaging the insertion aid shown in FIG. 3;
- (7) FIGS. 7 and 8 are cross-sectional views showing another nozzle of the present disclosure;
- (8) FIG. 9 is a flow chart illustrating one embodiment of a method for forming a sleeved catheter;
- (9) FIGS. 10 and 11 are perspective views of a system for delivering and/or withdrawing hydration medium from sleeved catheters;
- (10) FIG. 11a is a perspective view of an alternative to the delivery system;
- (11) FIG. 11b and 11c are perspective views of another alternative to the delivery system;
- (12) FIGS. 11d and 11e are perspective views of another alternative to the delivery system;
- (13) FIG. 11f is a perspective view of another alternative delivery system;
- (14) FIGS. 12-18 are views of insertion aids that include one or more openings or passages for injecting hydration medium into the interior cavity of the sleeve;
- (15) FIGS. 19 and 20 are views of drainage members that includes one or more openings or passages for injecting hydration medium into the interior cavity of the sleeve;
- (16) FIG. 21 is a perspective view of one embodiment of a catheter assembly in accordance with the present disclosure;
- (17) FIG. 22 is a perspective view of another embodiment of a catheter assembly in accordance with the present disclosure;
- (18) FIG. 23 is a cross-sectional view of the catheter assembly of FIG. 22;
- (19) FIG. 24 is a cross-sectional view of the catheter assembly of FIG. 22;
- (20) FIG. 25 is a plan view of another embodiment of a catheter assembly in accordance with the present disclosure;
- (21) FIG. 26 is a plan view of the catheter assembly of FIG. 25 showing the inner surface of the sleeve;
- (22) FIG. 27 is a cross-sectional view of the catheter assembly of FIG. 25;
- (23) FIGS. 28-30 are schematic views showing another embodiment of a catheter assembly in accordance with the present disclosure;

- (24) FIG. **31** is a perspective view of one embodiment of a hydration medium delivery mandrel in accordance with the present disclosure;
- (25) FIGS. **32** and **33** are partial cross-sectional view showing the hydration medium mandrel within a catheter assembly; and
- (26) FIG. **34** is a schematic view of one embodiment of a method of making a hydrophilic sleeved catheter assembly in accordance with the present disclosure;
- (27) FIG. **35** is a perspective view of one embodiment of a system and method of making a hydrophilic catheter assembly; and
- (28) FIG. **36** is a perspective view of another embodiment of a catheter assembly.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

(29) The embodiments disclosed herein are for the purpose of providing a description of the present subject matter, and it is understood that the subject matter may be embodied in various other forms and combinations not shown in detail. Therefore, specific embodiments and features disclosed herein are not to be interpreted as limiting the subject matter as defined in the accompanying claims.

(30) The present disclosure is directed to methods of making a sleeved hydrophilic urinary catheter product wherein the sleeved catheter is contained in an outer package and the urinary catheter is ready-to-use right out of the outer package. That is, while in the package, the hydrophilic outer surface of the catheter tube within the interior cavity of the sleeve is in a hydrated/activated state, so that the catheter is ready-to-use right out of the package.

(31) FIG. **1** illustrates one embodiment of a catheter assembly **10** in accordance with present disclosure. The catheter assembly **10** includes an elongated catheter tube **12** having a proximal end portion **14** and a distal end portion **16**. The proximal end portion **14** of the catheter tube **12** is suitable for insertion into a lumen or a passageway of the body, such as the urethra. The proximal end portion **14** may include drainage holes or eyelets **18** for draining urine from the bladder. A drainage member **20** may be associated with the distal end portion **16** of the catheter tube **12**. The catheter tube **12** includes an outer hydrophilic surface that becomes lubricious when hydrated or activated. The outer surface may be, for example, any suitable hydrophilic coating.

(32) The catheter assembly **10** also includes a sleeve **22**, which may be a protective or barrier sleeve that has a proximal end portion **24** and a distal end portion **26**. The sleeve **22** surrounds at least a portion of the catheter tube **12** to separate and enclose the portion of the catheter tube **12** from the outside environment. In other words, the protective sleeve **22** defines an interior cavity in which the catheter tube **12** may be located. In one embodiment, the sleeve **22** extends over the length of the catheter tube **12**. Optionally, an insertion aid **28** may be located at the proximal end portion **24** of the sleeve **22**. When an insertion aid **28** is present, the proximal end portion **24** of the sleeve **22** may be attached to a barrel or stem **30** of the insertion aid **28**, by for example, welding or adhesive. The distal end portion **26** of the sleeve **22** may be attached to the drainage member **20** or the distal end of the catheter tube **12**. An insertion aid may be used with any of the catheter assemblies disclosed herein.

(33) The sleeve **22** and any of the other sleeves disclosed herein may be made of a flexible material which may be vapor permeable or vapor impermeable, depending on the desired use and packaging. The material of the sleeve **22** may also be liquid impermeable. The sleeve **22** may be formed of any of a variety of thin, flexible polymeric film materials, such as polyethylene, plasticized PVC, or polypropylene, but elastomeric film materials such as polyurethane, and particularly elastomeric hydrogel materials, may be particularly suitable. The thickness of the film from which the sleeve **22** is formed may vary considerably depending on factors such as stretchability and flexibility of the material selected but, in general, the thickness may fall within the range of about 10 to 150 microns, preferably about 13 to 50 microns.

(34) Referring to FIGS. **1**, **2** and **3**, these figures illustrate exemplary embodiments of the insertion aids. In FIGS. **1** and **2**, the insertion aid **28** includes a proximal end portion **32** that defines an

introducer tip **34**. The introducer tip **34** has a proximal end aperture or opening **36** defined by one or more slits between one or more flexible petals **38**. The petals **38** may move, bend and/or resiliently deform from the generally closed aperture configuration shown in FIGS. **1** and **2** to an open aperture configuration (not shown) to allow for advancement of the catheter tube **12** therethrough. The distal end portion of the insertion aid **28** includes a cylindrical or barrel portion **30** that has an opening **40** for receiving the catheter tube **12**. The insertion aid **28** may also include an intermediate flange **42** that may contact the user about the urethra opening and act as a stop to limit the insertion of the introducer tip **34**.

(35) Turning to FIG. **3**, in this embodiment the insertion aid **28a** is a port **29a** that includes a flange **42a** surrounding an aperture or opening **34a**. The catheter tube **12** advances through opening **34a** for insertion into the urethra. The distal end portion of the port **29a** includes a cylindrical or barrel portion **30a** that has an opening **40a** for receiving the catheter tube **12**.

(36) Turning back to FIG. **1**, the insertion aid **28**, optionally, may be covered by a removable protective cap **44**. The removable protective cap **44** covers the insertion aid **28** and may protect the insertion aid **28** from contacting surfaces and objects prior to use.

(37) To use the catheter assembly **10**, the user opens and removes the catheter assembly **10** from an outer package (not shown). For example, the user opens the package and grasps the catheter tube **12** through the protective sleeve **22** to handle and manipulate the catheter assembly **10**. The user removes protective cap **44**, if one is present. If the catheter assembly **10** includes the insertion aid **28** shown in FIG. **2**, then the user inserts the introducer tip **34** into the urethra. If the catheter assembly **10** includes the insertion aid **28a** shown in FIG. **3**, then the user aligns the opening **34a** of the port **29a** with the urethral opening. The user then grasps the catheter tube **12** through the sleeve **22** and advances the catheter tube **12** through the insertion aid **28/28a** and into and through the urethra until the eyelets enter the bladder. If the catheter assembly **10** does not include an insertion aid, then the user grasps the catheter tube **12** through the sleeve **22** and advances the tip of the catheter tube **12** out of the open end of the sleeve **22** and into the urethra. When the eyelets enter the bladder, urine flows through the eyelets and catheter tube **12** to drain the bladder.

(38) In one method of making a sleeved hydrophilic catheter wherein the hydrophilic surface is in an activated or hydrated state, such as those described above, the method includes injecting or delivering a hydration medium into the interior cavity of the sleeve of the catheter assembly. While in the sleeve, the hydration medium contacts the hydrophilic surface of the catheter to at least partially hydrate or activate the hydrophilic surface, and in one embodiment, fully hydrate the hydrophilic surface. Optionally, the hydration medium dwells within the sleeve for a selected time period, which may be sufficient to partially or fully hydrate/activate the hydrophilic surface. Also optionally, an amount of hydration medium may be selectively withdrawn or extracted from the interior cavity of the sleeve. For example, some or all of the loose hydration medium may be withdrawn from the sleeve. The hydration medium within the hydrophilic surface will substantially remain within the hydrophilic coating and within the sleeve.

(39) The hydration medium may be a liquid, foam or a gel. For example, the hydration may be liquid water or an aqueous solution or any other suitable liquid hydration medium. In one embodiment, the hydration medium may be an aqueous solution that includes water, glycerol and, optionally, other additives.

(40) Optionally, the hydration medium may be a hydration foam that includes a liquid containing a mass of gas bubbles on or in the liquid. In one embodiment, the hydration foam medium includes, among other components, a liquid, a surfactant and gas. The liquid may be water or an aqueous solution. The surfactant may be any suitable foaming agent or surface tension reducing agent, such as sodium methyl cocoyl taurate or the like. The gas may be any suitable gas, such as ambient air, carbon dioxide, nitrogen, etc. The gas may be homogenized with the liquid to form a foam. When the hydration medium is a hydration foam, the hydration medium may be foamed and then delivered into the sleeve. Alternatively, the hydration medium may be foamed at the same time as it

is delivered into the sleeve, or may be foamed after it is delivered into the sleeve.

(41) In another embodiment, the hydration medium may be a water based gel. The gel based hydration medium may have a dual function, firstly hydrates hydrophilic coating and secondly protects retention of water. In one embodiment, the gel may be one that liquefies or becomes less viscous when exposed to radiation and may supplement hydration and lubriciousness of hydrophilic coating. For example, the gel may be a gellan gum based gel that is injected into the sleeve as a gel and then liquefies, breakdowns or becomes less viscous when the catheter assembly is exposed to sterilizing radiation, such as e-beam or gamma radiation. In one embodiment the gel may be a gel that includes 1.5 wt %-2 wt % of gellan gum, 1 wt % glycerol and 97 wt %-97.5 wt % of water.

(42) As will be discussed in more detail below, the hydration medium (liquid or gel) may have an elevated temperature during injection into the interior cavity of the sleeve. For example, the hydration medium may be at a temperature between 15° C.-70° C. In another embodiment, the hydration medium may be at a temperature between 40° C.-70° C. during injection. Any hydration medium not withdrawn from the sleeve and remaining in the sleeve after the withdrawal step may be allowed to cool to ambient temperatures. Injecting the hydration medium at an elevated temperature may assist in injecting and/or withdrawing the hydration medium. Additionally, injecting a hydration medium at an elevated temperature may lessen the time it takes for the hydration medium to hydrate/activate the hydrophilic surface of the catheter.

(43) When the hydration medium is a gel, the gel may be injected into the sleeve as a hot gel solution at an elevated temperature, as discussed above. The hot gel solution may partially or substantially hydrate the hydrophilic coating of the catheter tube. Optionally, a selected amount of the hot gel solution may be withdrawn. Alternatively, the method may not include a withdrawal step. The gel in the sleeve or remaining in the sleeve after a withdrawal step may cool to ambient temperatures (e.g., about 23° C. or below). When the gel cools, it may form a thin gel coating, such as a hydrogel coating, at least partially covering, and preferably substantially covering, the partially or substantially hydrated hydrophilic surface of the catheter tube. Additionally, there may be surplus deposits of gel located within the sleeve. Such gel deposits may be gel that is in the sleeve but not covering the catheter. Depending on the gel used, the gel may not hydrate the hydrophilic surface of the catheter while in the gel state, at least partially hydrate the hydrophilic surface of the catheter while in the gel state, or fully hydrate the hydrophilic surface of the catheter while in the gel state. Furthermore, the gel may be a gel that liquefies or becomes less viscous when the catheter assembly is exposed to sterilizing radiation. For example, after the gel injection step and optional withdrawal step, the gel may be covering the hydrophilic surface of the catheter and/or may otherwise be located in the sleeve. The catheter assembly is then placed in a package. The package may then be exposed to sterilizing radiation wherein the gel liquefies or becomes less viscous.

(44) Turning now to FIG. 4, this figure provides a schematic representation of a fill method that includes an injection system 52 for delivering hydration medium into the sleeve 22. Optionally, the system 52 may also have the capabilities of withdrawing hydration medium from the sleeve. The catheter assembly 10 may be docked or otherwise operatively connected to a hydration medium injection system or machine 52. The hydration medium injection and system 52 may include a source of hydration medium 54, which could be a reservoir or tank containing an amount of hydration medium 56. The system may include a conduit 58, one end 60 of which is connected to the source of hydration medium 54, and the end 62 of which is configured to be connected or docked to the catheter assembly 10 so that hydration medium 54 can be injected or delivered into and, optionally withdrawn from, the interior cavity of the sleeve 22. For example, the end 62 of the conduit 58 may include a nozzle 64 configured to be releasably connectable/docked to the sleeve 22 or the insertion aid 28, if one is present. The system also includes a pump or metering valves or other element 66 for moving/pumping hydration medium 56 so as to inject hydration medium into the sleeve 22. The pump or other element 66 may also move/pump hydration medium 56 so as to

withdraw or extract hydration medium **56** from the sleeve **22** and back into hydration source **54**. The injection and withdrawal pumps/metering valves/elements may be the same or different elements. For example, the system may include a reversible pump that injects and withdraws hydration medium, or the system may include one pump for injecting medium and another pump for withdrawing medium.

(45) As discussed above, the method of forming the sleeved activated hydrophilic catheter may include, injecting a hydration medium into the interior cavity of the sleeve, wherein the hydration medium comes into contact with the outer hydrophilic surface of the catheter tube. Optionally, allowing the hydration medium to dwell within the sleeve for a desired dwell time. Then, withdrawing at least some of the hydration medium from the interior cavity of the sleeve and placing the sleeve having the catheter tube therein in an outer package.

(46) Referring to FIGS. **4** and **9**, there is shown and described one exemplary embodiment of forming the sleeved hydrophilic catheter shown in FIG. **1**. If a protective cap **44** is present, the method includes removing protective cap **44** from the insertion aid **28**. The nozzle **64** of the injection system is then docked or connected to the insertion aid **28**. If the insertion aid **28** includes an introducer tip **34**, the nozzle **64** may dock to the catheter assembly **10** by inserting the introducer tip **34** into the nozzle **64**, as shown in FIG. **5**. The inner diameter of the nozzle **64** may have a size that generally corresponds to the outer diameter of the introducer tip **34**. As shown in FIGS. **7** and **8**, the nozzle **64a** may be a universal nozzle that may be configured to receive different sized introducer tips. For example, the inner surface of the nozzle **64a** may be stepped or include multiple steps **68a** or may have a generally cone shaped configuration for engaging and docking with different sized introducer tips **34a**, e.g., introducer tips having different sized diameters. Referring to FIG. **6**, when the insertion aid **28a** is a port **29a**, the nozzle **64b** may also be configured to dock to the port **29a**.

(47) Turning back to FIGS. **4** and **9**, after the nozzle **64** is docked, hydration medium **56** is injected from the nozzle **64** through the insertion aid **28** and into the sleeve **22** wherein the hydration medium contacts the hydrophilic surface of the catheter tube **12**. As mentioned above, the hydration medium may be injected at an elevated temperature. Furthermore, when an insertion aid **28** is present, the hydration medium **56** may be injected through the aperture **36** defined by slits and petals **38** (FIG. **2**). The petals **38** may resiliently deform to allow passage of the hydration medium **56** and/or the slits may have dimensions that allow for passage of hydration medium **56**.

(48) Optionally, the injected hydration medium **56** dwells in the sleeve **22** for a desired amount of time. For example, the dwell time may be a time that is sufficient for the hydration medium **56** to fully activate or fully hydrate the hydrophilic outer surface of the catheter tube **12**. In some embodiments, it may be desirable for the dwell time to be sufficient to partially hydrate or partially activate the hydrophilic outer surface of the catheter tube. Optionally, at least some of the hydration medium **56** is then withdrawn from the sleeve **22** through the insertion aid **28**. In one embodiment, substantially all of the loose hydration medium **56** is withdrawn from the sleeve **22**. That is, the hydration medium **56** that is loose within the sleeve **22** is withdrawn while the hydration medium that has been absorbed by the hydrophilic surface of the catheter tube **12** remains within the hydrophilic surface and the sleeve. In other embodiments, some of the loose hydration medium **56** is withdrawn and some is left in the sleeve **22**.

(49) After the hydration medium **56** has been withdrawn, the nozzle **64** is undocked and the protective cap **44** is placed on or refitted onto the insertion aid **28**. The catheter assembly **10** is then placed within an outer package (not shown) and the package is sealed. The outer package may then be submitted to sterilizing by e-beam or gamma radiation.

(50) In one embodiment, the outer package may be made of a gas impermeable and liquid impermeable material, such as a polymer and aluminum laminate. Furthermore, the package may be of the type that has a vapor atmosphere or 100% relative humidity within the seal package. For example, the package may include therein a water compartment that is at least partial defined by a

vapor permeable, liquid impermeable material. The water within the compartment may produce a water vapor that permeates through the vapor permeable, liquid impermeable material to create and/or maintain a hydration environment within the package. Additionally, when the catheter assembly is placed in a package having a vapor atmosphere, the sleeve may be vapor permeable to allow vapor to come into contact with the partially or substantially hydrated hydrophilic surface of the catheter tube. This may assist in maintaining the hydrophilic surface in an activated or hydrated state during storage and distribution. Alternatively, when the sleeve is made from a liquid and gas impermeable material and the interior cavity of the sleeve is sealed off, the outer package may be made from a gas permeable material.

(51) FIGS. **10** and **11** illustrate a portion of an injection system **52a** which may be employed to inject hydration medium into and, optionally withdraw hydration medium from, the interior cavity of the sleeve **22**. The system **52a** may include a holder **70** configured to accept and hold a plurality of catheter assemblies **10**. In the illustrated embodiment, the holder **70** includes a plurality of cavities for accepting and holding the catheter assemblies **10**. The holder **70** may include a body having separable front and rear body portions **72** and **74**, which separate to receive the catheter assemblies **10** into the respective cavities. The front and rear body portions **72** and **74** come together to orientate and hold the catheter assemblies **10** in place. After the catheter assemblies **10** are located in the holder **70**, the nozzles **76** may be docked or connected to the catheter assemblies **10**. In the illustrated embodiment, the nozzles **76** connect or dock to the introducer tips **34** of the catheter assemblies **10**. The system **52a** then injects hydration medium into the sleeve **22** and, optionally withdraws hydration medium from the sleeve, similar to that discussed above.

(52) FIG. **11a** illustrates alternative features of the system **52a**. The system **52a** may have compression members **71** that compress the introducer tips **34** of the insertion aids to place the apertures **36** of the insertion aids into the open configuration prior to or after the nozzle **76** (FIG. **11**) is attached. In the illustrated embodiment, the compression members **71** include a body **75** and a stem **77**. The stems **77** are located and slide in tracks **79** of the body portions **72** and **74**. In the illustrate embodiment, the track may be T-shaped and the stems **77** may include a corresponding T-shape. Opposed pairs of compression members **71** are positioned on either side of the introducer tip **34** and the opposed compression members **71** move toward each other to compress the introducer tip **34** and move the apertures **36** into the open configuration.

(53) FIG. **11b** and **11c** illustrate an alternative system **52b**. The system **52b** includes support members **71b** located on at least one of the front and rear bodies **72** and **74** of the holder **70**. The support members **71b** support the introducer tips **34** during delivery of the hydration medium. In the illustrate embodiment, the support members **71b** engage the introducer tips **34** when the holder is moved to the closed configuration. A nozzle or needle **76b** may then engage the introducer tips **34** to deliver the hydration medium.

(54) FIGS. **11d** and **11 e** illustrate another alternative system **52c**. The system **52c** includes a clamp **81** that clamps the catheter tubes **12** through the sleeves **22** and moves the catheter tubes **12** proximally and out of the introducer tips **34**. The delivery members, such as needles or nozzles **76b**, dock with the distal end of the catheter tubes **12**. The camp **81** is then moved proximally to move the catheter tubes **12** and the needles **76b** into the aperture **36** of the introducer tips **34**, and the hydration medium may be delivered through the tips **34** and in to the sleeve **22**.

(55) FIG. **11f** illustrates another alternative feature of the delivery systems. The delivery system may include disbursement members **83** that disburse the hydration fluid within the sleeve **22**. In the illustrated embodiment, the disbursement members **83** include opposed rollers that contact the sleeve **22** and roll along the sleeve **22** to disburse the hydration medium therein.

(56) FIGS. **12-20** illustrate exemplary embodiments of insertion aids and drainage members that have one or more openings or passageways therein for injecting and/or withdrawing hydration medium from the interior cavity of the sleeve **22**. The insertion aids and drainage members may include an opening(s) or a passageway(s) that communicates with the interior cavity of the sleeve

22. The opening(s) or passageway(s) may be used to inject and/or withdraw hydration medium from the interior cavity of the sleeve. For example, the openings and passageways may be configured to receive a device for injecting and withdrawing hydration medium, such as a nozzle or a needle. Additionally, any of the openings or passageways disclosed herein that are in communication with the interior cavity of the sleeve, whether they are in the insertion aid, drainage member or sleeve, may also serve as a vent that vents fluid from the sleeve when the sleeve is collapsed during use.

(57) FIGS. **12-18** illustrate exemplary embodiments of insertion aids that have a structure similar to that of insertion aid **28** described above. The insertion aids shown in these figures also include an opening for injecting and withdrawing a hydration medium. Turning first to FIG. **12**, in this embodiment, the insertion aid **28b** includes an opening **80b** in the barrel portion for injecting hydration medium into and withdrawing hydration medium from the interior cavity of the sleeve **22**. In FIG. **13**, the insertion aid **28c** includes an opening **80c** in the barrel portion for injection and/or withdrawal of hydration medium wherein the opening **80c** is covered by a portion of the sleeve **22**. In this embodiment, a needle or nozzle may penetrate the portion of the sleeve to access the opening **80c**. In FIG. **14**, the insertion aid **28d** includes petals **82d** in the barrel portion that form a cross-shaped deformable slit **80d**. In FIG. **15**, the insertion aid **28e** includes a deformable slit **80e** in the barrel portion, and in FIG. **16**, the insertion aid **28f** includes a Y-shaped slit in the barrel portion. In each of these embodiments, a nozzle may be pressed against the slits **80d**, **80e** and **80f** to open the slits or a needle may penetrate the slits to inject and/or withdraw hydration medium. Additionally, similar to FIG. **13**, a portion of the sleeve **22** may cover the slits **80d**, **80e** and **80f**, wherein the portion of the sleeve **22** must be penetrated to access the slit. FIG. **17** illustrates another embodiment wherein the insertion aid **28g** includes a U-shaped slit **80g** that is accessible by a needle or nozzle **84** to inject a hydration medium **86**, and optionally withdraw the hydration medium. In FIG. **18**, the insertion aid **28h** includes a plurality of axial passageways **80h** for injecting and/or withdrawing hydration medium.

(58) FIGS. **19** and **20** illustrate drainage members **20a** and **20b** that include one or more passageways that are configured for injecting hydration medium into and/or withdrawing hydration medium from the interior cavity of the sleeve **22**. Turning first to FIG. **19**, the drainage member **20a** includes passageways **90a** having an opening **92a** external of the sleeve **22** and another opening **94a** within the interior cavity of the sleeve **22**. The external opening **92a** is configured to receive a device, such as a nozzle or needle **84a**, for injecting hydration medium **86a** into and/or withdrawing hydration medium **86a** from the sleeve **22**. Optionally, the passageways **90a** may be sealed or closed after the delivery of the hydration medium. For example, caps or plugs could be placed in the passageways **90a**. In one embodiment, a slideable sealing ring **91a** could be associated with the drainage member **20a**. After the hydration medium has been delivered, the ring **91a** can be slide over the passageways **9ab** to close or seal them. Referring to FIG. **20**, the drainage member **20b** includes one or more axial passageways **90b** for injecting hydration medium into and/or withdrawing hydration medium from the sleeve **22**.

(59) Another embodiment is that the nozzle may include an atomizer which converts hydration medium into small droplets which can be sprayed onto hydrophilic coating surface to induce hydration in sleeved cavity.

(60) Another embodiment is that the hydration fluid is delivered as steam through the nozzle in sleeved cavity and onto hydrophilic coating surface to induce hydration. In another embodiment, the hydration fluid is delivered as a foam, for example, a foaming agent such as air or nitrogen could be added to the hydration fluid.

(61) FIG. **21** illustrates another embodiment of a catheter assembly **100**. The catheter assembly **100** includes catheter **102**, a sleeve **104**, an insertion aid **106** and a removal protective cap **108**. In the illustrated embodiment, the insertion aid **106** includes an insertion tip **110** wherein the tip includes a proximal end opening **112** defined by slits **114** and petals **116**. The insertion aid **106** may also be

any of the insertion aids disclosed herein or any other suitable insertion aid.

(62) The protective cap **108**, optionally, includes a ring **118** at the proximal end thereof wherein the ring **118** may be sized and configured to be grasped by a user. For example, a user may grasp ring **118** or place a finger in the ring **118** to remove it from the catheter assembly **110**. When a ring **118** is present, a portion **119** of the band of the ring **118** may include an opening **120** therethrough. The opening **120** in the band may be defined by a rim **121** in the portion **119**. Additionally, the body of the cap **108** may include a proximal end opening **122** which may be aligned with the opening **112** of the insertion aid **106**. Furthermore, the opening **120** in the portion **119** of the band of ring **118** may be aligned with the opening **122** of the cap **108**. In the illustrated embodiment opening **122** of the cap **108** is defined by a rim **124**. In other embodiments, the opening **122** may be defined by one or more slits (not shown) and/or petals, or it may be an opening that can be opened and closed. For example, the opening may be opened and closed by flexible material or the opening may be covered or plugged.

(63) Optionally, the cap **108** may include a funnel member **123** that extends from opening **122** and into the opening **112** of the insertion aid **106**. The delivery device **128** may be inserted through the funnel member **123**, or the hydration fluid may be delivered through the funnel member **123**.

(64) To fill and/or withdraw hydration medium from the interior cavity of the sleeve **104**, a hydration medium delivery device, such as nozzle or needle **128**, is inserted through opening **120** in ring **118**, when one is present, and into opening **122** of the cap **108**. The nozzle or needle **128** is then inserted into opening **112** of the insertion aid **106** and hydration medium is injected into and/or withdrawn from the interior cavity of the sleeve **104**.

(65) FIGS. **22-24** illustrate another embodiment of a catheter assembly **130**. The catheter assembly **130** includes a catheter **132** having drainage member **134** associated with the distal end portion of the catheter **132**. The catheter assembly **130** also includes a sleeve **136** that may be used for the injection and/or withdrawal of hydration medium. The sleeve **136** may be a two-part sleeve that includes a proximal segment **138** and a distal segment **140**. The proximal and distal segments **138** and **140** define one or more openings or passageways **139** therebetween that may be used to injection and/or withdraw hydration mediums. The proximal segment **138** of the sleeve **136** extends proximally to cover the catheter tube and, optionally, includes an introducer tip (not shown) associated therewith. The distal segment **140** of the sleeve may be attached to the drainage member **134**.

(66) As shown in FIGS. **22** and **24**, the proximal segment **138** and the distal segment **140** of sleeve **136** may be attached to each other along the sides of each segment. For example, sides **142** of distal segment **140** of sleeve **136** may be attached to side **144** of proximal segment **138** of sleeve **136**. In the illustrated embodiment, openings **139** are located on either side of the attachments. In other embodiments, the proximal and distal segments **138** and **140** may include a plurality of attachments spaced apart about the segments or there may be only one attachment. The proximal and distal sleeve segments **138** and **140** may be attached to each other in any suitable manner, such as be heat sealing, solvent bonding or adhesive.

(67) To inject hydration medium into and/or withdraw hydration medium from the interior cavity of sleeve **136**, a hydration medium delivery device, such as a nozzle or needle (not shown), may be inserted into or access opening(s) or passageway(s) **139**.

(68) FIGS. **25-27** illustrate another embodiment of a catheter assembly **150** wherein in the sleeve **152** includes openings or passageways **158** (FIG. **27**) that may be used to inject and/or withdraw hydration medium to and from the interior cavity of the sleeve **152**. The assembly **150** also includes a catheter **154** and a drainage member **156**. FIG. **25** illustrates the catheter assembly **150** in the assembled configuration and FIG. **26** illustrates the catheter assembly **150** with the sleeve **152** separated so as to show the inner surface **160** of the sleeve **152**. Referring to FIG. **26**, a ply or layer of material **162** is attached to the inner surface **160** of the sleeve **152** such that opening or passageway **158** (FIG. **27**) is defined between the inner surface **160** of the sleeve and the layer of

material **162**. For example, the layer **162** may be attached to the inner surface **160** of the sleeve **152** along side-edges **164a** and **164b**. The layer **162** may be attached to the sleeve **152** in any suitable manner, such as be heat sealing, solvent bonding or adhesive. The proximal edge **166a** and the distal edge **166b** are not attached to the inner surface **160** such that opening or passageway **158** (FIG. 27) is defined between the layer **162** and sleeve **152**.

(69) As shown in FIG. 27, the sleeve **152** may be attached to the drainage member **156**. For example, a portion of the sleeve **168** and layer **162** may be attached to the drainage member **156**. To inject hydration medium into and/or withdraw hydration medium from the interior cavity of sleeve **152**, a hydration medium delivery device, such as a nozzle or needle (not shown), may be inserted into or access opening or passageway **158**. Optionally, a one-way valve **157** may be located within or associated with passageway **158**. The one-way valve may be attached to or molded with the drainage member **156**. The hydration medium may be delivered through the one-way valve **157**. The one-way valve will prevent hydration medium from exiting the sleeve **152**.

(70) FIGS. 28-30 illustrate another embodiment of a catheter assembly **170** wherein in the sleeve **172** includes openings or passageways **174** that may be used to inject and/or withdraw hydration medium from the interior cavity of the sleeve **172**. The sleeve **172** includes one or more openings or passageways **174** therethrough. In the illustrated embodiment, the opening or passageway **174** is in the distal end of the sleeve. In other embodiments, the opening or passageway **174** may be in other locations of the sleeve, such as in the middle or proximal end of the sleeve.

(71) To inject hydration medium into and/or withdraw hydration medium from the interior cavity of sleeve **172**, a hydration medium delivery device, such as a nozzle or needle **176**, may be inserted into or access opening or passageway **174**. After the hydration medium has been injected and/or withdrawn and the nozzle or needle has been removed from the opening or passageway **174**, a patch or layer **178** is placed over the opening or passageway **174** and attached to the sleeve **172** to cover opening or passageway **174**. The patch or layer **178** may be attached by adhesive or any other suitable manner. Additionally, the patch **178** may be in the form of a flap wherein one end of the patch is connected to the sleeve **172**. When the patch is a flap, the flap initially covers the passageway **174**. The flap is peeled back for insertion of hydration medium, and then placed back over the passageway **174**.

(72) In an alternative embodiment, a needle may pierce the sleeve **172** and hydration medium is delivered through the needle. The needle may be removed and a patch **178** may be placed over the needle hole. The piercing and patch applying may be done with the aid of a location device.

(73) FIG. 31 illustrate one embodiment of a mandrel or tube **180** that may be employed to inject hydration medium into and/or withdraw hydration medium from the interior cavity of a sleeve of a catheter assembly. The mandrel includes a proximal end portion **182** and a distal end portion **184**. The distal end portion **184** includes an opening **186** for receiving a hydration medium. In the illustrated embodiment, the distal end portion **184** includes a hub **185** that may be connected to a hydration medium source. The proximal end portion **128** includes one or more openings **188** that are in fluid communication with opening **186** in the distal end portion **184**. For example, the mandrel **180** may include a conduit or passageway (not shown) extending from the opening **186** to the one or more openings **188**. In the illustrated embodiment, the opening **188** in the proximal end portion **182** includes a plurality of pin-holes. In other embodiments, the proximal end portion **182** may include a single opening.

(74) Referring to FIGS. 32 and 33, the mandrel **180** may be inserted and advanced into the drainage member **192** of a catheter **194**. The proximal end portion **182** and holes **188** may be aligned with eyelets or openings **196** of the catheter **194**. Hydration medium may then be advanced through the mandrel **180** and out of the holes **188**. The hydration medium then exits through the eyelets **196** of the catheter and into the interior cavity of the sleeve **198**. Optionally, hydration medium may be extracted from the interior cavity of the sleeve **198**. For example, the hydration medium may be extracted through the eyelets **196** of the catheter **194** and through the holes **188** in the mandrel **180**.

(75) FIG. 34 illustrates another method of forming a sleeved hydrophilic catheter assembly. In this method, a hydration medium **90** may be applied to the hydrophilic surface of the catheter tube **12**. For example, the hydration medium **90** may be applied by dip coating, spraying, steaming or any other suitable application process. In the illustrated embodiment, the hydration medium **90** is in a hydration medium reservoir **92** and the catheter tube **12** is dipped into the hydration medium **90**. Optionally, the hydration medium may be at an elevated temperature as described above. For example, the hydration medium **90** may be a liquid, aqueous solution, foam or gel at an elevated temperature. When the catheter tube **12** is dipped in the hydration medium **90**, the catheter tube **12** may be allowed to dwell in the hydration medium **90** for a selected period of time. The selected period of time may be sufficient to fully or partially hydrate/activate the hydrophilic surface of the catheter. The catheter **12** is then removed from the hydration medium **90**. When the hydration medium is a gel, the gel may form a gel coating that covers at least a portion of the hydrophilic surface of the catheter tube and preferably covers the entire hydrophilic surface. The sleeve **22** is then assembled around the catheter tube **12** and, optionally, an inserter aid and protective cap is assembled with the sleeve. The sleeve may be any of those discussed above or any other suitable sleeve. The catheter assembly is placed in a package, such as any of those discussed above. For example, the package may include a vapor hydration atmosphere and the sleeve may be vapor permeable, as discussed above. The package may then be submitted to sterilizing radiation, such as e-beam or gamma radiation. When a gel is used as the hydration medium, the gel may be of the type that liquefies or becomes less viscous when exposed to radiation.

(76) FIG. 35 illustrates another embodiment of a system and method for applying a hydration medium to a catheter. The system may include a sprayer **200** that sprays hydration medium on to the hydrophilic surface of the catheter tubes **12**.

(77) FIG. 36 illustrates another hydrophilic catheter product and method of forming the same. The catheter product includes a top sheet **204** and a bottom sheet **206** that are sealed or attached together along their peripheral edges **208** and **210**, respectively, to form the sleeve surrounding the catheter tube **12**. At least one of the inner surfaces of the top sheet **204** and the bottom sheet **206** is coated with a hydration medium **212**, such as a hydration gel that releases hydration liquid. The hydration medium may be applied in any sufficient manner. For example, the hydration medium may be rolled, sprayed or brushed on the sheet. After the top and bottom sheets are attached to each other, the gel is activated to release the hydration liquid.

(78) It will be understood that the embodiments described above are illustrative of some of the applications of the principles of the present subject matter. Numerous modifications may be made by those skilled in the art without departing from the spirit and scope of the claimed subject matter, including those combinations of features that are individually disclosed or claimed herein. For these reasons, the scope hereof is not limited to the above description but is as set forth in the following claims, and it is understood that claims may be directed to the features hereof, including as combinations of features that are individually disclosed or claimed herein.

Claims

1. A method of making a ready-to-use urinary catheter product, wherein the product includes a catheter assembly with a sleeve defining an inner cavity and a urinary catheter having a catheter tube located within the inner cavity of the sleeve, the catheter tube having an outer hydrophilic surface, an insertion aid at a distal end of the sleeve configured for passage of the catheter tube therethrough during catheterization, a protective cap and the protective cap includes a passageway, the method comprising: inserting the hydration medium delivery device through the passageway of the protective cap and engaging a hydration medium delivery device with an aperture of the insertion aid; delivering a hydration medium from the hydration medium delivery device, wherein the hydration medium flows from the delivery device through the insertion aid and immediately

- into the inner cavity of the sleeve where the hydration medium is loose and hydrates the outer hydrophilic surface of the catheter tube; and placing the catheter assembly with the hydrated hydrophilic surface in an outer package.
2. The method of claim 1, further including withdrawing at least some of the hydration medium from the inner cavity of the sleeve.
 3. The method of claim 1, wherein the insertion aid defines the aperture in communication with the inner cavity of the sleeve, and wherein the step of engaging the hydration medium delivery device with the insertion aid comprises engaging the hydration medium delivery device with the aperture and the step of delivering hydration medium through the insertion aid and into the inner cavity of the sleeve comprises delivering hydration medium through the aperture in the insertion aid.
 4. The method of claim 3, wherein the aperture is moveable between a closed configuration and an open configuration.
 5. The method of claim 4, further including moving the aperture into the open configuration prior to delivering the hydration medium through the insertion aid.
 6. The method of claim 4, wherein delivering the hydration medium moves the aperture from the closed configuration to the open configuration.
 7. The method of claim 4, wherein the insertion aid includes a plurality of petals defining the aperture.
 8. The method of claim 7, wherein the petals move to define the open configuration of the aperture.
 9. The method of claim 3, wherein the insertion aid has a proximal end and a distal end, the insertion aid including an introducer tip at the proximal end thereof, and the aperture being located in the introducer tip.
 10. The method of claim 9, wherein the aperture is configured to have the catheter tube advanced therethrough during catheterization.
 11. The method of claim 3, wherein the insertion aid includes a barrel portion and the aperture is located in the barrel portion.
 12. The method of claim 11, wherein a portion of the sleeve covers the aperture.
 13. The method of claim 1, wherein the catheter product includes a protective cap covering the insertion aid.
 14. The method of claim 1, wherein the passageway of the protective cap includes a funnel-shaped member in communication with the aperture of the insertion aid.
 15. The method of claim 1, wherein the protective cap includes a grasping ring defined by a band and the band includes an opening therethrough for allowing the passage of the hydration medium delivery device.
 16. The method of claim 1, wherein engaging the hydration medium delivery device with the insertion aid comprises docking the insertion aid to a nozzle of a hydration medium delivery system.
 17. The method of claim 2, wherein withdrawing at least some of the hydration medium comprises withdrawing at least some of the hydration medium through the insertion aid and the hydration medium delivery device.
 18. The method of claim 1, wherein the hydration medium comprises a hydration foam.
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